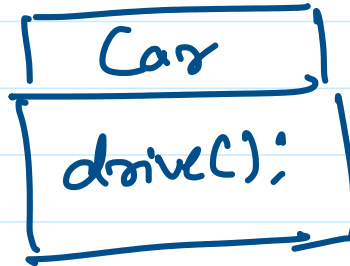


This Design Pattern is also for specific use-case.

Let understand the problem



There are various types of car

Sedan

Hatchback

SUV

⋮

⏟
m

Various types of Engine

Petrol

Electric.

Diesel.

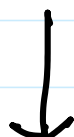
⏟
n

Now you can see lot of Permutation & Combination is possible

Total $m * n$ classes will be created

↳ So it is getting exponentially ↑

This is called "CLASS EXPLOSION"



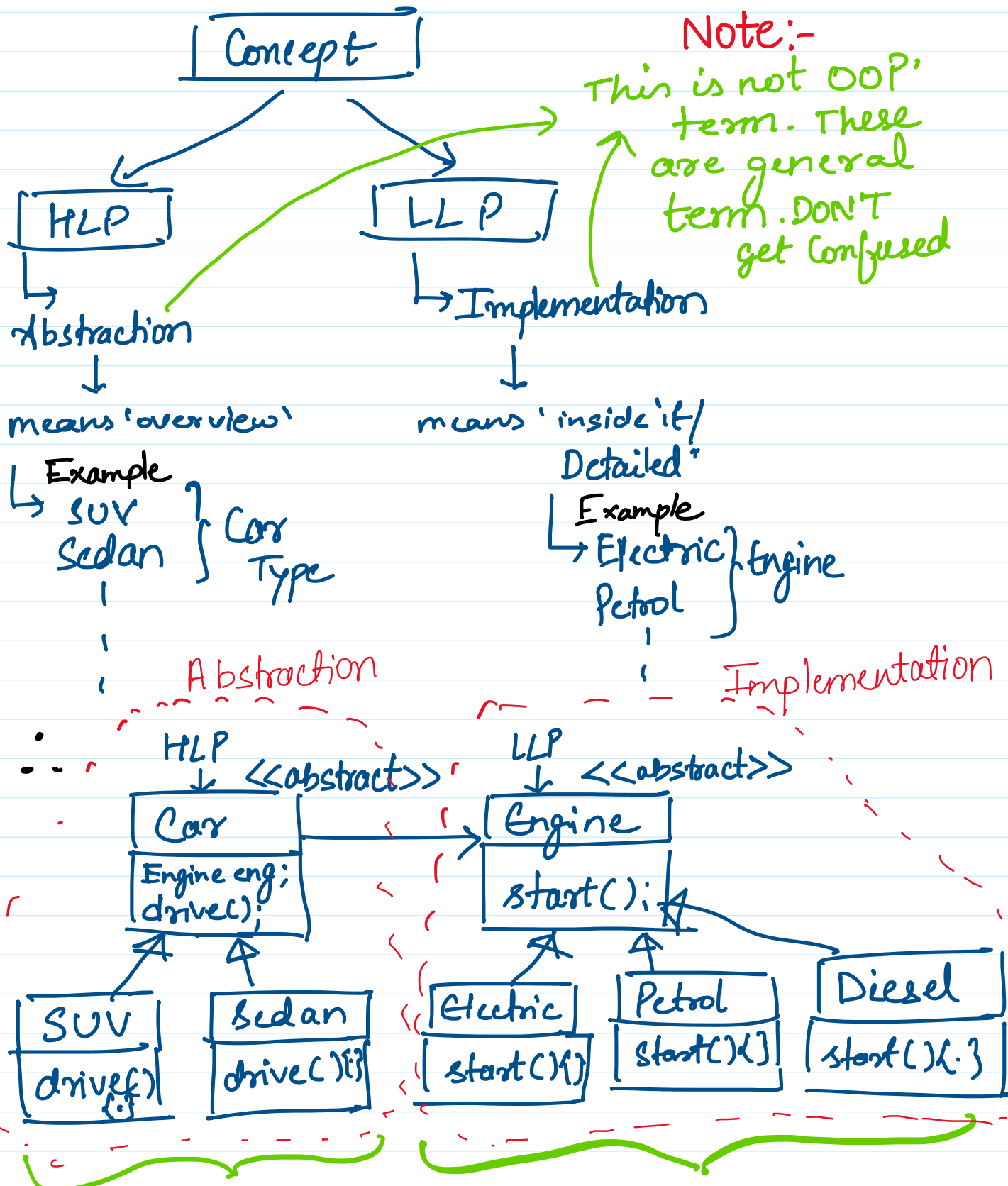
↓

It says we have created lot of classes to implement a feature in our application
↳ This leads to:-

1. Complicated System
2. Code Reuse ↓
3. lot of classes
4. Difficult to understand relationship b/w classes.

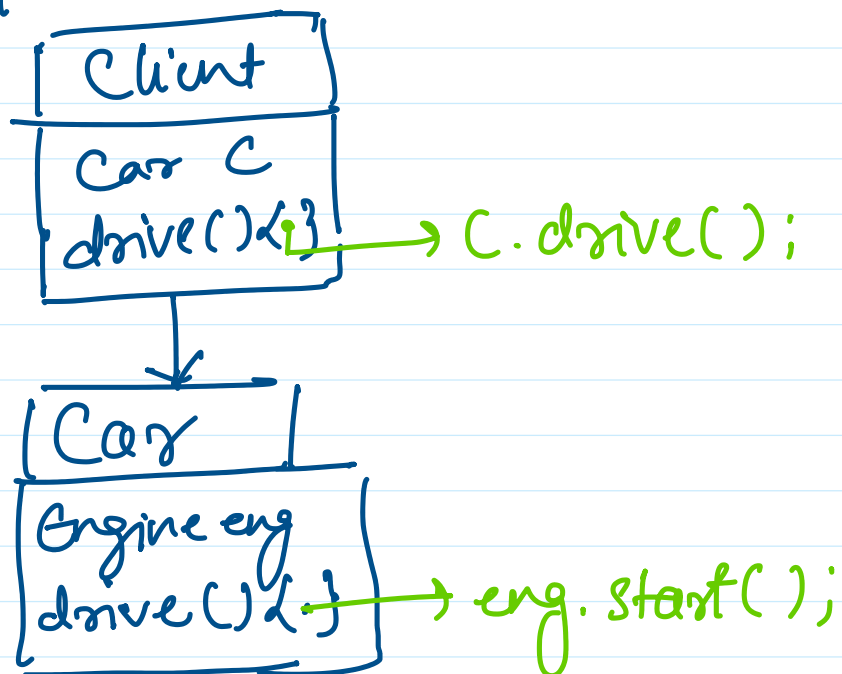
What Bridge Design Pattern Says:-

Whatever your concept is Break it into 'Low Level Part' & 'High Level Part'

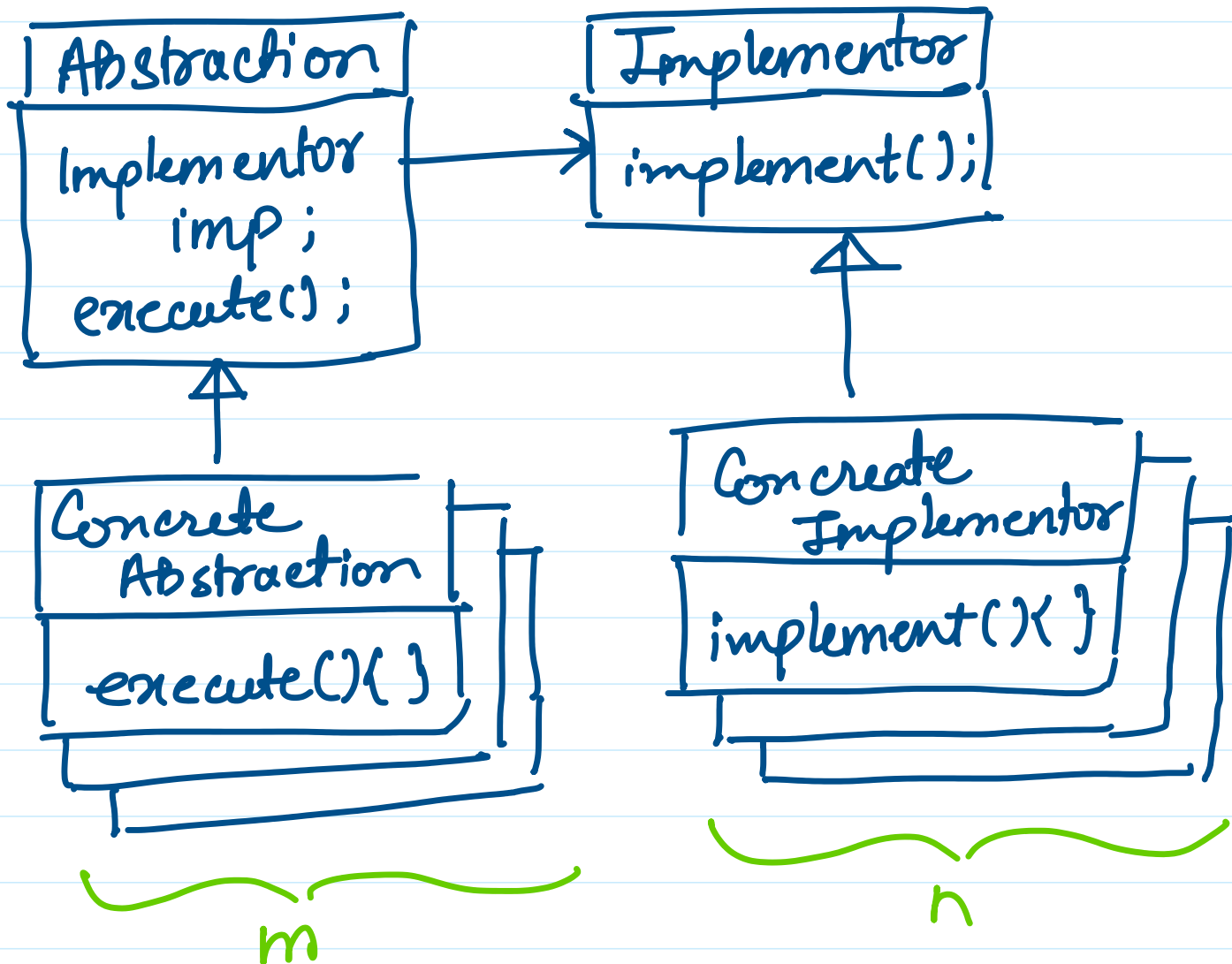


Now we don't need $m \times n$ class
Now just we need $m+n$ class
B/c in 'eng' we can inject any type
of Engine.

Here Client
look like
this :-



Standard UML Diagram



Standard Definition

Bridge Decouples an abstractions from its implementations, so that both can vary independently.

- Abstractions: High level layer (Car)

- Implementations : how Level Layer (Engine)

Code → see Notes....

Strategy V/s Bridge Pattern

Both Pattern UML look same if you think deeply

Here, Car $\rightarrow$ Robot Class (Strategy)
Notes

Engine $\rightarrow$ features like
talkable, walkable etc.
(Strategy Notes)

Robot has-a reference
of feature class
Same
here
also

Car has Reference of Engine Class.

Then what is the difference?

INTENT

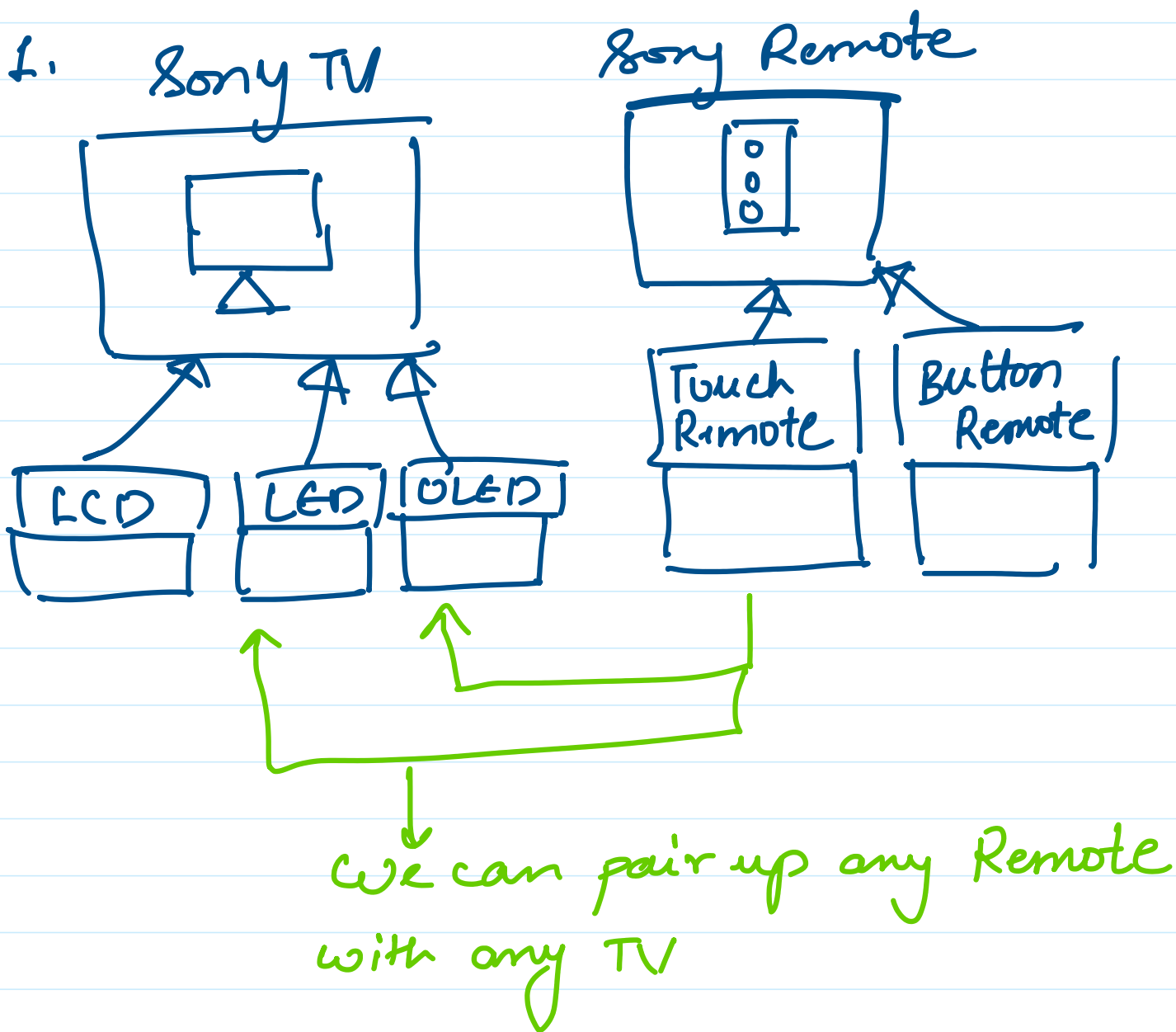
Strategy - There are multiple algorithms
and our client can use it dynamically.
 $\downarrow$

2

A single lowercase letter 'l' is written in blue ink on a piece of white paper with horizontal blue lines. The letter is formed with a single stroke, starting from the middle line, going down to the bottom line, and then curving slightly to the right at the bottom.

A single lowercase letter 'l' is written in blue ink on a piece of white paper with horizontal blue lines. The letter is formed with a single stroke, starting from the middle line, going down to the bottom line, and then curving slightly to the right at the bottom.

Real World Use Cases



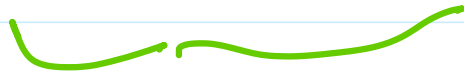
2. There are some GUI which looks different in different OS so,

GUI

dropdown

Radio Button

Text Box



Abstraction

OS

windows

MAC OS

LINUX



Implementations